Can Deniz Bezek

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Profile

I am a recent Ph.D. graduate specializing in medical and computational imaging, deep learning, signal processing, and inverse problems. I develop end-to-end imaging and machine learning pipelines spanning data acquisition, preprocessing, image formation, model development, performance evaluation, quantitative validation, and clinical applications. My doctoral research has focused on tomographic reconstruction of speed-of-sound images from ultrasound data, combining imaging physics, analytical modeling, and physics-informed and deep-learning-based approaches to solve the underlying inverse problem. I am motivated to bring this expertise in imaging science and computational methods to medical imaging R&D.

Education

Ph.D. in Information Technology — November 2021 – September 2026
Uppsala University — Uppsala, Sweden

- Thesis: Pulse-Echo Speed-of-Sound Reconstruction for Quantitative Ultrasound Imaging
- Supervisor: Prof. Orcun Goksel | Co-supervisor: Prof. Ingela Nyström

M.Sc. in Electrical and Electronics Engineering (CGPA: 4/4) — 2019 – 2021
Middle East Technical University (METU) — Ankara, Türkiye

- Thesis: Deep Learning-Based Unrolled Reconstruction Methods for Computational Imaging
- Supervisor: Assoc. Prof. Figen S. Oktem

B.Sc. in Electrical and Electronics Engineering (CGPA: 3.88/4) — 2015 – 2019
Middle East Technical University — Ankara, Türkiye

Technical Skills

Programming: Python, MATLAB, C
Deep Learning & AI: PyTorch, TensorFlow, model-based deep learning, unrolled reconstruction, network pruning
Imaging & Signal Processing: Image formation, tomographic reconstruction, computational imaging, inverse problems, imaging physics and modeling, ultrasound imaging, beamforming, research prototyping, statistical and digital signal processing
Tools: Linux, Git, Weights & Biases, LabVIEW, LaTeX
Languages: Turkish (native), English (fluent), German (fluent), Swedish (beginner)

Academic and Professional Experience

Doctoral Researcher — November 2021 – September 2026
Uppsala University — Uppsala, Sweden

- Developed end-to-end ultrasound imaging pipelines for speed-of-sound estimation, spanning optimized data acquisition, RF signal processing, tomographic reconstruction via analytical and generalizable physics-informed deep learning methods, and prototyping on clinical ultrasound systems
- Validated the developed algorithms for clinical applications in breast cancer assessment, breast density classification, and liver fat quantification
- Assisted in teaching the Medical Informatics course and mentoring incoming Ph.D. students

Visiting Researcher — June 2024 – August 2024
Robotics and Control Laboratory, The University of British Columbia — Vancouver, Canada

- Developed speed-of-sound imaging methods using conventional ultrasound transducers and laser-diode photoacoustic sources
- Contributed to developing a speed-of-sound imaging pipeline for a prostate cancer study

Research and Teaching Assistant — February 2020 – November 2021
Middle East Technical University — Ankara, Türkiye

- Developed deep learning-based reconstruction methods for multispectral and compressive spectral imaging

- Assisted in teaching Vector Space Methods in Signal Processing, Probability and Random Variables, and Real-Time Applications of Digital Signal Processing courses

Project Assistant November 2018 – June 2019
Arcelik A.S. Ankara, Türkiye

- Developed multicast DNS functionality on a microcontroller for IoT applications in household appliances

Research Intern June 2018 – September 2018
Medizinische Informationstechnik (MedIT) Aachen, Germany

- Developed capacitive electrocardiogram (ECG) mock-up prototype
- Simulated and analyzed ballistocardiographic coupling into capacitive ECG

Awards, Certificates & Honors

IVA's List for Research Impact | Selected as one of 30 Swedish research projects with strong potential for societal benefit through innovation, commercialization, and collaboration, 2026

IEEE International Ultrasonics Symposium Student Travel Grant | Awarded, on a competitive basis, 2026 & 2023

Dubai Future Solutions – Prototypes for Humanity | Selected as one of 100 showcased projects out of 3,300+ applications across 100+ countries; travel, accommodation, and subsistence covered, 2025

Anna Maria Lundins Scholarship | Travel grant to present at IEEE International Ultrasonics Symposium, 2025

IFMBE-MTF Best Poster Award | Best poster award at Medicinteknikdagarna, 2024

Liljewalch Travel Scholarships | Travel grant for a research visit to The University of British Columbia, 2024

METU Course Performance Award | Graduate student with the highest CGPA in EEE Department, 2020

Korea Advanced Institute of Science and Technology (KAIST) | Travel grant to join KAIST EE Camp, 2019

Bulent Kerim Altay Award | Awarded three times for achieving a 4/4 GPA, 2018 – 2019

TUBITAK (Scientific and Technological Research Council of Türkiye) | Scholarship for M.Sc. studies, 2019 – 2021

METU | Listed in Dean's High Honor Roll for all semesters, 2015 – 2019

Deutsches Sprachdiplom (DSDII) | German proficiency at level C1 (except listening B2)

IELTS | Overall score 7.5

Journal Publications

1. **C. D. Bezek***, M. Farkas*, D. Schweizer, R. A. Kubik-Huch, and O. Goksel, "Breast Density Assessment via Quantitative Sound-Speed Measurement Using Conventional Ultrasound Transducers", **European Radiology**, 2025.

2. D. Schweizer, M. Farkas, **C. D. Bezek**, A. Potempa, C. Leo, R. A. Kubik-Huch, and O. Goksel, "Pulse-Echo Imaging of Breast Speed-of-Sound as a Potential Biomarker for Breast Cancer", **Ultrasound in Medicine & Biology**, 2025.

3. S. Laguna, L. Zhang, **C. D. Bezek**, M. Farkas, D. Schweizer, R. A. Kubik-Huch, and O. Goksel, "Uncertainty Estimation for Trust Attribution to Speed-of-Sound Reconstruction with Variational Networks", **International Journal of Computer Assisted Radiology and Surgery**, 2025.

4. **C. D. Bezek**, M. Haas, R. Rau, and O. Goksel, "Learning the Imaging Model of Speed-of-Sound Reconstruction via a Convolutional Formulation", **IEEE Transactions on Medical Imaging**, 2024.

5. D. Schweizer, R. Rau, **C. D. Bezek**, R. A. Kubik-Huch, and O. Goksel, "Robust Imaging of Speed-of-Sound Using Virtual Source Transmission", **IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control**, 2023.

6. **C. D. Bezek** and O. Goksel, "Analytical Estimation of Beamforming Speed-of-Sound Using Transmission Geometry", **Ultrasonics**, 2023.

7. F. S. Oktem, O. F. Kar, **C. D. Bezek**, and F. Kamalabadi, "High-resolution Multi-spectral Imaging with Diffractive Lenses and Learned Reconstruction", **IEEE Transactions on Computational Imaging**, 2021.

Conference Proceedings

1. **C. D. Bezek** and O. Goksel, "DenOiS: Dual-Domain Denoising of Observation and Solution in Ultrasound Image Reconstruction", **Reconstruction and Imaging Motion Estimation (RIME) Workshop, in conjunction with MICCAI**, *accepted*, 2026.
2. **C. D. Bezek** and O. Goksel, "Windowed Sound-Speed Prediction by Extending Beamforming-Based Global Estimators", **IEEE International Ultrasonics Symposium**, 2025.
3. **C. D. Bezek**, P. Koudelka, R. Denkin, and O. Goksel, "Sound Speed Approximation Using B-mode Image Alignment with Steered Focused Transmits", **IEEE International Ultrasonics Symposium**, 2025.
4. L. Zhu, **C. D. Bezek**, and O. Goksel, "FGGP: Fixed-Rate Gradient-First Gradual Pruning", **Scandinavian Conference on Image Analysis**, 2025.
5. **C. D. Bezek***, H. Moradi*, R. Rohling, S. Salcudean, and O. Goksel, "Towards Speed-of-Sound Imaging with Conventional Ultrasound Transducers Using Laser Diode Photoacoustic", **SPIE Medical Imaging**, 2025.
6. **C. D. Bezek** and O. Goksel, "Model-Based Speed-of-Sound Reconstruction via Interpretable Pruned Priors", **IEEE International Ultrasonics Symposium**, 2024.
7. **C. D. Bezek**, M. Bilgin, L. Zhang, and O. Goksel, "Global Speed-of-Sound Prediction Using Transmission Geometry", **IEEE International Ultrasonics Symposium**, 2022.
8. **C. D. Bezek** and F. S. Oktem, "Unrolling-Based Deep Reconstruction for Compressive Spectral Imaging", **Computational Optical Sensing and Imaging**, 2021.
9. I. Manisali*, R. M. Cam*, **C. D. Bezek***, and F. S. Oktem, "Deep CNN Prior Based Image Reconstruction for Multispectral Imaging", **28th Signal Processing and Communications Applications Conference**, 2020.
10. D. U. Uguz, P. Weidener, **C. D. Bezek**, T. Wang, S. Leonhardt and C. H. Antink, "Ballistocardiographic Coupling of Triboelectric Charges into Capacitive ECG", **IEEE International Symposium on Medical Measurements and Applications**, 2019.

Manuscripts Under Review and In Preparation

1. D. Schweizer, **C. D. Bezek**, and O. Goksel, "Fast Alternating Radial Beamforming for Speed-of-Sound Imaging", *under review*, 2026.
2. **C. D. Bezek** and O. Goksel, "Pulse-Echo Ultrasound Methods for Speed-of-Sound Estimation: A Review", *under review*, 2026.

Conference Talks

1. DenOiS: Dual-Domain Denoising of Observation and Solution in Ultrasound Image Reconstruction, **Reconstruction and Imaging Motion Estimation Workshop, MICCAI**, *accepted for presentation*, 2026.
2. Pulse-Echo Sound-Speed Imaging Using Conditional Diffusion Models, **IEEE International Ultrasonics Symposium**, *accepted for presentation*, 2026.
3. Reconstruction-Aware Measurement Refinement for Imaging Sound-Speed, **IEEE International Ultrasonics Symposium**, *accepted for presentation*, 2026.
4. Pulse-Echo Speed-of-Sound Estimation and Its Applications, **The Artimino Conference on Medical Ultrasound Technology**, Lyon, France, 2025.
5. Reconstruction of Ultrasound-Speed Maps with a Learned Imaging Model, **Swedish Symposium on Image Analysis**, Stockholm, Sweden, 2025.
6. Speed-of-Sound as a Novel Tissue Characterization Method, **International Tissue Elasticity Conference**, Lyon, France, 2024.
7. Pulse-Echo Speed-of-Sound as Imaging Biomarker for Breast Density: Virtual Source Acquisitions for In-Vivo Application, **IEEE International Ultrasonics Symposium**, Montreal, Canada, 2023.
8. Speed-of-Sound as a Novel Ultrasound Imaging Biomarker for Breast Cancer and Density, **Medicinteknikdagarna**, Stockholm, Sweden, 2023.

9. Model-Based Deep Learning of Ultrasound Beamforming, **Swedish Symposium on Image Analysis**, Norrköping, Sweden, 2023.
10. Global Speed-of-Sound Prediction Using Transmission Geometry, **IEEE International Ultrasonics Symposium**, Venice, Italy, 2022.
11. Mean Speed-of-Sound Estimation Using Geometric Disparities, **Swedish Symposium on Image Analysis**, virtual, 2022.

Poster Presentations

1. Windowed Sound-Speed Prediction by Extending Beamforming-Based Global Estimators, **IEEE International Ultrasonics Symposium**, Utrecht, Netherlands, 2025.
2. Sound Speed Approximation Using B-mode Image Alignment with Steered Focused Transmits, **IEEE International Ultrasonics Symposium**, Utrecht, Netherlands, 2025.
3. FGGP: Fixed-Rate Gradient-First Gradual Pruning, **Scandinavian Conference on Image Analysis**, Reykjavík, Iceland, 2025.
4. Speed-of-Sound as a Novel Quantitative Imaging and Characterization Method, **Medicinteknikdagarna**, Gothenburg, Sweden, 2024.
5. Model-Based Speed-of-Sound Reconstruction via Interpretable Pruned Priors, **IEEE International Ultrasonics Symposium**, virtual, 2024.
6. Motion Sensitivity of Transmit Sequences for Pulse-Echo Mapping of Sound Speed based on Apparent Speckle Shifts, **IEEE International Ultrasonics Symposium**, Montreal, Canada, 2023.
7. Sound-Speed Reconstruction with Learned Kernels based on a Convolutional Formulation of Sound-Speed and Speckle-Shift Relation, **IEEE International Ultrasonics Symposium**, Montreal, Canada, 2023.

Supervised Students

1. Haben Hadush Gebreyowhans, Implicit Neural Representations for Speed of Sound Imaging, Master's Thesis, 2026.
2. Zezheng Zhang, Diffusion models for ultrasound imaging of sound-speed, Master's Thesis, 2025.
3. Paul Koudelka, Interactive Speed of Sound Measurement with Ultrasound Using Computational Methods, Master's Thesis, 2025.
4. Lingkai Zhu, Edge Pruning Strategies in Neural Networks, Master's Thesis, 2024.
5. Ema Duljkovic, Model-Based Deep Learning for Ultrasound Imaging of Sound Speed, Master's Thesis, 2024.

Professional Service

Journal Reviewer

- IEEE Transactions on Computational Imaging
- IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control

Volunteer Activities

IEEE METU Student Branch May 2016 – June 2017

Led a team of 50+ students for a three-day career event attended by over 3,000 visitors.

Editor of tr.motorsport.com November 2015 – February 2016